

A Minor Project
on
GROUND STATION FOR TRACKING LORA SATELLITES
SUBMITTED IN THE PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE
AWARD OF THE DEGREE OF
BACHELOR OF TECHNOLOGY
(Electronics & Communication Engineering)



JAN-MAY 2026

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Candidate's Declaration

We hereby certify that the work which is being presented in the Project entitled “GROUND STATION FOR TRACKING LoRa SATELLITES” “ GURNOOR KAUR, GURPUNEET SINGH HUNJAN” in partial fulfilment of requirements for the award of degree of B.Tech. (Electronics and Communication Engineering) submitted to the Department of Electronics and Communication Engineering at GURU NANAK DEV ENGINEERING COLLEGE, LUDHIANA is an authentic record of my own work carried out during a period from _____ to _____. The matter presented in this project has not been submitted by us or anybody else in any other University / Institute for the award of B.Tech Degree.

Signature of the Student

This is to certify that the above statement made by the candidate is correct to the best of my own knowledge.

(Signature of Project Guide/Guides)

The Major Project Viva–Voce Examination of _____ has been held on _____ and accepted.

Signature of Internal Examiner

Signature of External Examiner

Abstract

The LoRa-based Satellite Ground Station is a mini project designed to develop a low-cost and efficient system for tracking and receiving telemetry data from LoRa-enabled Low Earth Orbit (LEO) satellites. The system is implemented using an ESP32 microcontroller, which acts as the central processing unit and is programmed through the Arduino IDE. It integrates components such as the RA-02 (SX1278) LoRa module for long-range wireless communication, an OLED display for real-time data visualization, and a suitable antenna system for signal reception. This project provides a practical understanding of satellite communication, embedded systems, and Low Power Wide Area Network (LPWAN) technologies.

The working of the proposed system is based on the reception of LoRa-modulated signals transmitted by satellites orbiting the Earth. The ground station is capable of automatically configuring and tuning itself to the required satellite frequencies to ensure reliable communication. It receives telemetry data such as battery voltage, temperature, and system health parameters, which are then decoded and processed by the ESP32. The processed data is displayed locally and can also be transmitted to cloud-based platforms like TinyGS for real-time monitoring and long-term data analysis. The implemented logic includes frequency management, packet decoding, signal validation, and continuous data acquisition, making the system robust and scalable.

This project is highly beneficial for students as it enhances practical knowledge in the domains of microcontrollers, wireless communication, and satellite-based IoT systems. It also strengthens concepts related to programming logic, hardware interfacing, and real-time data processing. The LoRa-based Satellite Ground Station demonstrates the potential of integrating embedded systems with modern communication technologies to develop cost-effective and scalable solutions, making it highly suitable for B.Tech 3rd-year academic projects as well as research-oriented applications.

Acknowledgement

We would like to express our sincere gratitude to Guru Nanak Dev Engineering College (GNDEC) for providing us the opportunity to undertake the minor project titled “Ground Station for Tracking LoRa Satellites”. This project has been a valuable experience that helped us gain practical knowledge and technical skills beyond classroom learning.

We are deeply thankful to our project guides, Dr. Munish Rattan and Dr. Shivmanmeet Singh, for his guidance, continuous support, and valuable suggestions throughout the project. His mentorship significantly improved our understanding of satellite communication, embedded systems, and real-world implementation.

We also extend our gratitude to our institution, Guru Nanak Dev Engineering College, Ludhiana, and the faculty of the Electronics and Communication Engineering department for their constant support and resources.

Finally, we thank our family and friends for their motivation and encouragement, which kept us focused and dedicated throughout the project.

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CHAPTER 1

INTRODUCTION & PROJECT FORMULATION

1.1 Overview

The LoRa-based Satellite Ground Station is an interactive embedded system project developed using ESP32 and LoRa communication modules. The system is designed to receive and monitor telemetry data transmitted by LoRa-enabled Low Earth Orbit (LEO) satellites. The main objective of this project is to integrate concepts of embedded systems, wireless communication, and satellite technology into a practical and real-time application. It helps in understanding how microcontrollers can be used for long-range communication and satellite data processing.

This project uses an ESP32 microcontroller as the core processing unit, which manages signal reception, data decoding, and communication with external platforms. A LoRa module such as RA-02 (SX1278) is used to receive satellite signals, while an OLED display or cloud dashboard is used to visualize the received telemetry data. The reception of signals, frequency tuning, packet decoding, and data transmission are programmed using embedded C/C++ in the Arduino IDE. By integrating hardware and software components, the system demonstrates the practical implementation of satellite communication and IoT-based monitoring.

The LoRa-based Satellite Ground Station is highly useful as it provides hands-on experience in wireless communication, signal processing, and embedded system design. It enhances understanding of LPWAN technologies, real-time data acquisition, and cloud-based monitoring systems. Apart from its educational value, the project also demonstrates the application of LoRa in modern satellite-based IoT systems. Overall, this project serves as an effective learning model for communication and embedded system development.

1.2 Existing System

The existing system refers to the current methods used for satellite data reception and monitoring before the development of the proposed LoRa-based ground station. Traditionally, satellite communication systems require large, complex, and expensive

ground stations equipped with high-power transmitters, parabolic antennas, and licensed frequency bands. These systems are mainly used by government organizations, research institutions, and space agencies.

In the current system, satellite telemetry data is received through advanced communication infrastructure and processed using specialized software and hardware systems. These systems provide accurate and reliable data but are not easily accessible to students or small-scale projects due to their high cost and complexity. The user generally interacts with the system through software interfaces, without understanding the internal working of signal reception and processing at the hardware level.

Another existing approach includes internet-based satellite data platforms where users can access processed satellite information online. While these platforms are useful for observation, they do not provide hands-on experience in signal reception, decoding, or embedded system design.

From an academic perspective, the existing systems have several limitations. They do not provide practical exposure to low-cost communication technologies like LoRa or allow students to build and experiment with real hardware. There is minimal involvement in circuit design, hardware interfacing, and real-time signal processing. As a result, the learning remains theoretical rather than practical.

The major drawbacks of the existing system include:

- High cost and complex infrastructure
- Limited accessibility for students
- No practical exposure to hardware implementation
- Lack of understanding of real-time signal processing

- Dependency on pre-built systems and software
- Limited flexibility for experimentation and modification

Due to these limitations, there is a need for a cost-effective and simplified system that integrates both hardware and software for satellite communication. This requirement leads to the development of the proposed LoRa-based satellite ground station.

1.3 User requirement analysis

User requirement analysis involves identifying the expectations and needs of users from the proposed system. In the LoRa-based satellite ground station project, the user requires a reliable, easy-to-use, and efficient system for receiving and monitoring satellite data. The system should be capable of automatically detecting satellite signals and displaying meaningful telemetry data without complexity.

The user expects the system to receive satellite data such as battery status, temperature, and system health parameters accurately. The data should be clearly displayed on a screen or dashboard for easy understanding. The system should respond efficiently to signal reception and provide continuous monitoring without interruptions.

User Requirements:

- Reliable reception of LoRa satellite signals
- Clear display of telemetry data (battery, temperature, etc.)
- Automatic frequency tuning and configuration
- Real-time data processing and monitoring
- Simple and user-friendly interface
- Proper hardware and LoRa module interfacing
- Stable and accurate system performance

- Useful for learning satellite communication concepts
- Easy to demonstrate as a mini project

1.4 Feasibility Study

Feasibility of study is performed to determine whether the proposed project can be successfully implemented with available resources, time, and budget. For the LoRa-based satellite ground station, the feasibility includes technical, economic, and operational aspects.

Technical Feasibility

The required components such as ESP32, RA-02 LoRa module, OLED display, and antenna are easily available and widely used in IoT projects. The programming environment (Arduino IDE) is user-friendly and supports LoRa communication libraries. The implementation of signal reception, decoding, and data display can be achieved using standard embedded programming techniques. Therefore, the project is technically feasible.

Economic Feasibility:

The project is cost-effective as it uses low-cost components compared to traditional satellite communication systems. The overall budget is affordable for students, and the learning outcomes are significant. The benefits in terms of practical knowledge and project demonstration are much higher than the cost, making it economically feasible.

Operational Feasibility:

The system is simple to operate and requires minimal user intervention. Once configured, it can automatically receive and process satellite signals. The interface is easy to understand, and the system can be used for both educational and demonstration purposes. Hence, the project is operationally feasible.

Overall, the LoRa-based satellite ground station is technically, economically, and operationally feasible.

1.5 Objectives of Project

The main objective of this project is to design and develop a LoRa-based satellite ground station system that demonstrates the practical application of embedded systems and wireless communication technologies. The project aims to create a system capable of receiving, decoding, and monitoring satellite telemetry data in real time.

Objectives of the Project:

- To design and develop a LoRa-based satellite ground station using ESP32
- To receive telemetry data from LoRa-enabled LEO satellites
- To implement automatic frequency tuning and configuration
- To decode and display satellite data such as battery and temperature
- To integrate hardware components with embedded programming
- To understand the practical use of wireless communication systems
- To develop a low-cost and scalable communication system
- To demonstrate real-time data reception and monitoring
- To enhance knowledge of satellite communication and IoT systems

CHAPTER 2

PROJECT DESIGN

2.1 Project perspective

The LoRa-based Satellite Ground Station project is designed with the perspective of integrating satellite communication with embedded system learning. The primary aim of this project is to develop a low-cost and efficient ground station capable of receiving telemetry data from LoRa-enabled satellites. This project highlights how modern wireless communication technologies like LoRa can be utilized for long-range, low-power satellite data reception using minimal hardware resources.

From a technical perspective, the project focuses on the integration of hardware components such as the ESP32 microcontroller, RA-02 LoRa module, antenna system, and OLED display. The ESP32 acts as the central processing unit, handling signal reception, frequency tuning, data decoding, and communication with cloud platforms. The system is programmed to receive LoRa-modulated signals, process telemetry packets, and display meaningful information such as satellite health parameters, signal strength, and packet status.

From a user perspective, the system is designed to be simple, reliable, and easy to operate. Once configured, the ground station can automatically receive satellite signals without requiring continuous manual intervention. The display and dashboard provide clear visualization of received data, ensuring easy interpretation for users with basic technical knowledge.

From an educational perspective, the project enables students to understand advanced concepts such as satellite communication, LPWAN technology, signal processing, and IoT-

based monitoring. It also enhances practical skills in embedded system design, hardware interfacing, and real-time data acquisition.

From a future perspective, the project can be extended by adding features such as advanced antennas for improved reception, integration with cloud analytics platforms, multi-satellite tracking, and mobile-based monitoring applications. It can also be upgraded using high-performance microcontrollers and SDR-based systems for enhanced capability.

Overall, this project demonstrates the practical implementation of a low-cost satellite ground station and serves as a foundation for advanced communication and IoT-based systems.

2.2 Project functions

The LoRa-based Satellite Ground Station performs several key functions for real-time satellite tracking and data monitoring:

- **Signal Reception**

The system receives LoRa-modulated signals transmitted by LEO satellites using the RA-02 LoRa module and antenna system.

- **Frequency Configuration**

The system automatically tunes to required satellite frequencies to ensure reliable signal reception.

- **Data Decoding**

The received raw data packets are decoded into meaningful telemetry information such as battery voltage, temperature, and system health.

- **Real-Time Processing**

The ESP32 processes incoming data continuously and ensures proper handling of received packets.

- **Display Output**

Telemetry data is displayed on an OLED screen for real-time visualization.

- **Cloud Integration**

The system uploads received data to online platforms like Tiny GS for remote monitoring and long-term storage.

- **Signal Quality Monitoring**

Parameters such as RSSI, SNR, and packet success rate are monitored to evaluate communication performance.

- **Continuous Tracking**

The system continuously tracks multiple satellites without manual intervention.

- **System Initialization & Reset**

The system initializes automatically on startup and allows reset for reconfiguration if required.

2.3 User characteristics

The LoRa-based Satellite Ground Station is designed considering a diverse set of users with varying technical backgrounds and interaction capabilities. The key user characteristics are as follows:

- **Basic to Intermediate Technical Knowledge**

The primary users include students (B.Tech, Diploma, or hobbyists) who possess fundamental knowledge of electronics, embedded systems, and communication technologies. However, the system operation is designed to be simple and does not require deep expertise in satellite communication.

- **Familiarity with Digital Interfaces**

Users are expected to have basic familiarity with microcontrollers, communication modules, and simple interfaces such as displays or dashboards, enabling smooth interaction with the system.

- **Analytical and Observation Skills**

The system requires users to interpret telemetry data such as signal strength, packet information, and satellite parameters. Users should be capable of understanding real-time data output.

- **Age Group Diversity**

The system is suitable for a wide age group, typically ranging from students to young professionals. It can also be used by beginners under guidance due to its simple interface.

- **Learning-Oriented Users**

A significant category of users includes learners who aim to understand satellite communication, LoRa technology, and real-time embedded system implementation through practical application.

- **Technical and Research-Oriented Users**

Apart from educational purposes, the system also targets users interested in research, experimentation, and IoT-based satellite communication systems.

- **Adaptability to User Skill Levels**

The system supports flexibility in configuration and expansion, making it suitable for both beginners and advanced users by modifying parameters such as frequency settings and data processing.

- **Minimal Training Requirement**

Due to its user-friendly design and automated functionality, the system requires minimal training, making it easily accessible to new users.

- **Engagement and Interaction Preference**

Users are expected to prefer systems with real-time data feedback, including visual display and cloud-based monitoring, enhancing overall user experience.

2.4 Constraints

The design and implementation of the LoRa-based Satellite Ground Station are subject to several technical and practical constraints, which influence system performance and scalability:

- **Hardware Resource Limitations**

The ESP32 microcontroller and LoRa module have limited processing power, RAM, and memory, which restrict the complexity of data processing, signal decoding, and multitasking capabilities.

- **Signal Reception Constraints**

Satellite signals are weak and depend on factors such as antenna quality, positioning, and environmental conditions, which may affect reliable data reception.

- **Processing Speed and Real-Time Performance**

Real-time decoding of LoRa signals and continuous data handling can be challenging, especially when multiple satellites or high packet rates are involved.

- **Communication Range and Frequency Limitations**

The system operates within specific ISM frequency bands, which may limit flexibility and performance depending on regional regulations and interference.

- **Power Supply Constraints**

The system is typically powered using USB or battery sources, which imposes limitations on power consumption and continuous long-duration operation.

- **Scalability Restrictions**

Expanding the system to support multiple antennas, advanced tracking mechanisms, or large-scale networks is limited by hardware capabilities and system design.

- **Memory Constraints for Data Handling**

Storing large amounts of telemetry data locally is restricted due to limited onboard memory, requiring dependency on cloud platforms.

- **Wiring and Circuit Complexity**

Integration of multiple components such as LoRa modules, displays, and antennas increases circuit complexity, which may lead to signal noise or connection issues.

- **Network Dependency**

Cloud integration and remote monitoring depend on stable internet connectivity, which may not always be available.

- **Environmental and Physical Constraints**

External factors such as weather conditions, electromagnetic interference, and physical obstructions can affect signal quality and system performance.

- **Cost Constraints**

Although designed to be low-cost, budget limitations restrict the use of high-gain antennas, advanced RF components, and high-performance processing units.

2.5 Methodology

The development of the LoRa-based Satellite Ground Station follows a systematic hardware-software co-design approach to ensure efficient real-time performance and reliable system behaviour.

The process begins with requirement analysis, where key functionalities such as signal reception, data decoding, display output, and cloud communication are defined. Based on these requirements, suitable hardware components such as ESP32, LoRa module, OLED display, and antenna are selected.

In the hardware design phase, all components are interfaced properly with correct pin configurations, ensuring stable power supply and signal integrity. Special attention is given to antenna placement and wiring to reduce noise and improve signal reception.

On the software side, the system is developed using Embedded C/C++ in the Arduino IDE. The core logic includes signal reception, frequency tuning, packet decoding, and data display. LoRa libraries are used for communication, and display libraries are used for visualization.

After development, system integration is carried out to ensure smooth interaction between hardware and software. The system continuously monitors incoming signals and processes them in real time.

Testing and debugging are performed to verify system performance. Serial monitoring and real-time observation are used to detect and fix errors.

Finally, optimization techniques are applied to improve efficiency, reduce delays, and ensure stable operation. The system is validated to ensure accurate data reception and reliable performance.

2.6 Equipment and apparatus required

➤ Microcontroller Unit (Processing Core)

- ESP32

This is the central processing unit of the system. It executes the ground station firmware, processes incoming LoRa packets, handles automated frequency tuning, and manages Wi-Fi connectivity to upload data to the cloud servers.



Fig 1. ESP - 32

➤ Communication Module

- **RA-02 LoRa Module** This module uses the SX1278 chip to receive Chirp Spread Spectrum (CSS) modulated signals. It allows the station to intercept telemetry data sent from satellites over very long distances while consuming minimal power.



Fig 2. RA-02 LoRa Module

➤ Display Unit

• OLED

(0.95")

Used to visually represent the local system status, including network connectivity, current listening frequency, and real-time packet reception stats.

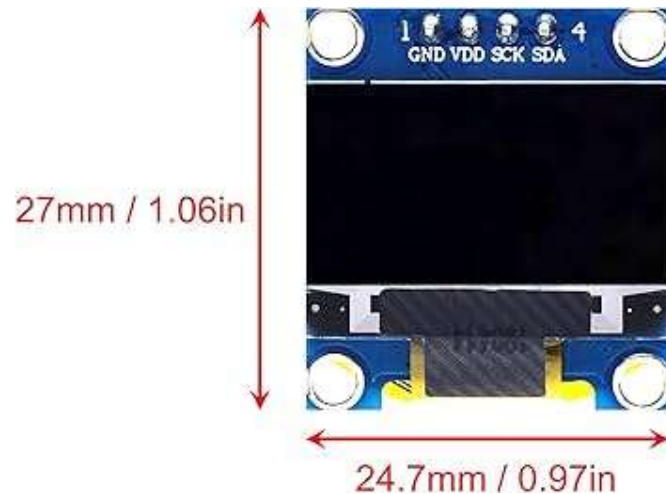


Fig 3. 0.95" OLED

➤ Power Regulation

• Buck (Step-Down) Converter

An electronic module used to step down the input voltage to a safe, regulated 5V DC level. It ensures that the ESP32 and LoRa modules receive stable power without overheating or short-circuiting.



Fig 3. Buck Converter

➤ Antenna System

- **Antenna & U.FL to SMA Patch Cord**

The antenna is crucial for capturing the weak RF signals transmitted by LEO satellites. The U.FL to SMA patch cord provides a secure, low-loss connection between the RA-02 module and the external antenna.



Fig.5 Antenna & U.FL to SMA Patch Cord

➤ Connecting Components

- **PCB and Data Cable**

- Printed Circuit Boards (PCBs) are used as the base structure to mount all components securely, providing a strong and organized surface. Data cables are used to supply power and flash firmware from the development laptop to the ESP32.

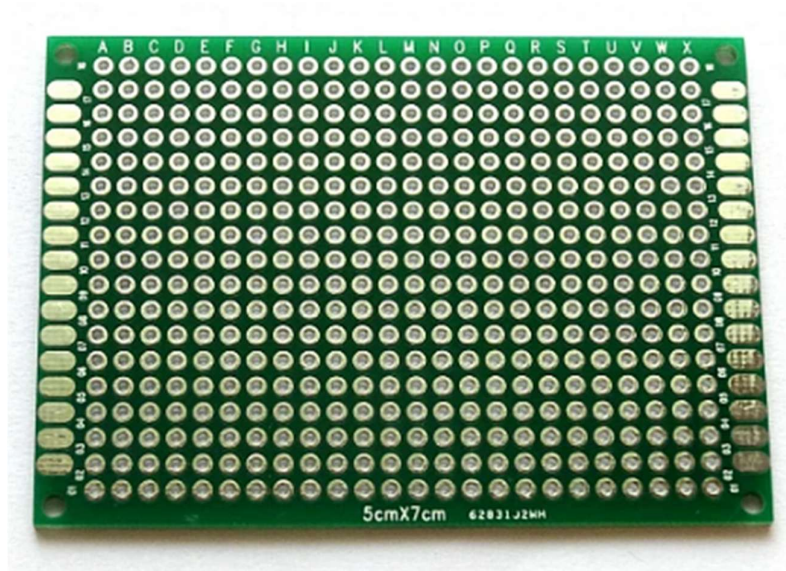


Fig.6 PCB

➤ PVC Box

A PVC box is used as the main protective enclosure to house and mount all components securely. It provides a strong, lightweight, and non-conductive environment for safely placing the ESP32 microcontroller, RA-02 LoRa module, OLED display, and buck converter, ensuring proper internal arrangement and protection from short circuits. The PVC box also helps in maintaining durability against external environmental factors and gives the ground station project a neat, professional, and organized appearance.



Fig.11 PVC Box

2.7 Assumptions and Dependencies

➤ Assumptions

The development and functioning of the LoRa-based Satellite Ground Station are based on the following assumptions:

- **Stable Power Supply**

It is assumed that the system receives a continuous and regulated power supply without fluctuations to ensure uninterrupted operation.

- **Proper Hardware Functionality**

All hardware components such as ESP32, LoRa module (RA-02), antenna, and display module are assumed to be in fully working condition without defects.

- **Correct Circuit Connections**

It is assumed that all interfacing connections are properly configured as per the circuit diagram with no loose or incorrect wiring.

- **Reliable Signal Availability**

It is assumed that LoRa-enabled satellites are transmitting signals and are within reception range during system operation.

- **Pre-Installed and Functional Software Environment**

The Arduino IDE and required LoRa and display libraries are assumed to be correctly installed and configured.

- **Sufficient Processing Capability**

The ESP32 microcontroller is assumed to be capable of handling signal reception, data decoding, and output display without performance issues.

- **Fixed Communication Parameters**

The communication settings such as frequency, bandwidth, and spreading factor are assumed to remain constant unless modified in the program.

➤ **Dependencies**

The system functionality depends on several hardware, software, and environmental factors:

- **Microcontroller Dependency**

The entire system operation depends on the ESP32 microcontroller. Any failure directly affects performance.

- **LoRa Module Dependency**

Signal reception depends on the proper functioning of the RA-02 LoRa module.

- **Antenna Dependency**

Signal strength and quality depend heavily on antenna design, placement, and orientation.

- **Display Module Dependency**

The visual output depends on the OLED display. Without it, real-time data visualization is not possible.

- **Power Supply Dependency**

Continuous operation depends on a stable power source. Any interruption may cause system reset or failure.

- **Software Dependency (Arduino IDE & Libraries)**

The system depends on properly compiled code and required libraries for LoRa communication and display control.

- **Network Dependency**

Cloud integration (e.g., TinyGS) depends on stable internet connectivity.

- **Hardware Interfacing Dependency**

Proper functioning depends on correct interfacing between all components and signal integrity.

- **Environmental Dependency**

External factors such as interference, obstacles, and weather conditions may affect signal reception and system performance.

2.8 Specific Requirements

The LoRa-based Satellite Ground Station has certain specific requirements that define what the system must achieve in terms of functionality, performance, and design. These requirements ensure that the system operates efficiently and provides accurate real-time satellite data monitoring.

- Functional Requirements

- The system should receive LoRa signals from satellites using a LoRa module.
- The system should decode received telemetry data into meaningful parameters.
- The system must display satellite data such as battery voltage, temperature, and signal strength.
- The system should support automatic frequency configuration for satellite

communication.

- The system should upload received data to a cloud platform (e.g., TinyGS).
- The system should continuously monitor and track satellite signals in real time.

- Hardware Requirements

- ESP32 microcontroller
- RA-02 LoRa module (SX1278)
- OLED display for data visualization
- Antenna for signal reception
- Jumper wires and PCB for connections
- Power supply (USB or battery source)

- Software Requirements

- Arduino IDE for coding and uploading program
- Embedded C/C++ programming language
- LoRa communication libraries and display libraries

- Performance Requirements

- The system should receive and process signals in real time with minimal delay.
- Data decoding and display should be accurate and stable.
- The system should operate continuously without data loss or interruption.
- Signal reception should be reliable under normal environmental conditions.

- Design Requirements

- The circuit should be simple and compact for easy implementation.
- The code should be modular for easy debugging and modification.
- The system should be energy efficient and suitable for long-duration operation.

CHAPTER 3

DEVELOPMENT AND IMPLEMENTATION

3.1 Introduction to Software

To program the LoRa-based Satellite Ground Station, a suitable development environment is required where the code can be written, compiled, and uploaded to the microcontroller. This is achieved using the **Arduino IDE**, which supports multiple operating systems such as Windows, Linux, and macOS.

The Arduino IDE is an Integrated Development Environment (IDE) used to write programs in Embedded C/C++ and upload them to microcontrollers like ESP32. It provides a simple interface for coding, compiling, debugging, and serial monitoring.

In this project, the Arduino IDE is used along with required libraries such as LoRa communication libraries and OLED display libraries. The program written in the IDE is known as a *sketch*, and it is saved with a “.ino” extension.

The software plays a critical role in handling LoRa signal reception, decoding satellite telemetry data, and displaying it on the OLED screen as well as transmitting it to cloud platforms like TinyGS.

3.2 Implementation using Screen Shots/Figures/Graphs



Fig.12 Components used

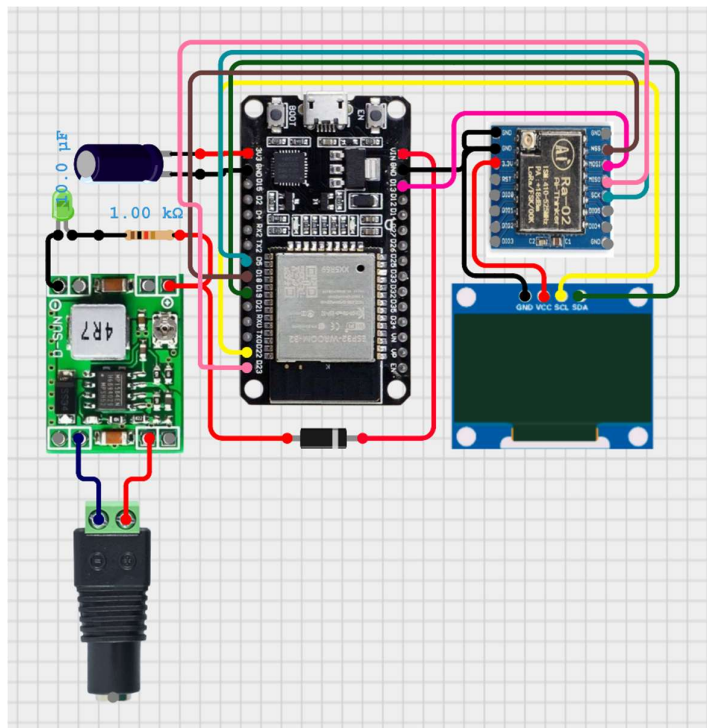


Fig.13 Circuit diagram

3.3 Testing & Verification

In the testing phase of the LoRa-based Satellite Ground Station, each hardware component such as the ESP32 microcontroller, LoRa module (RA-02), OLED display, and antenna is checked individually to ensure proper functioning. After verifying individual components, the complete system is integrated and the program is uploaded to observe real-time signal reception and data display. Key functions such as LoRa signal reception, data decoding, display output, and serial monitoring are tested to verify correct operation.

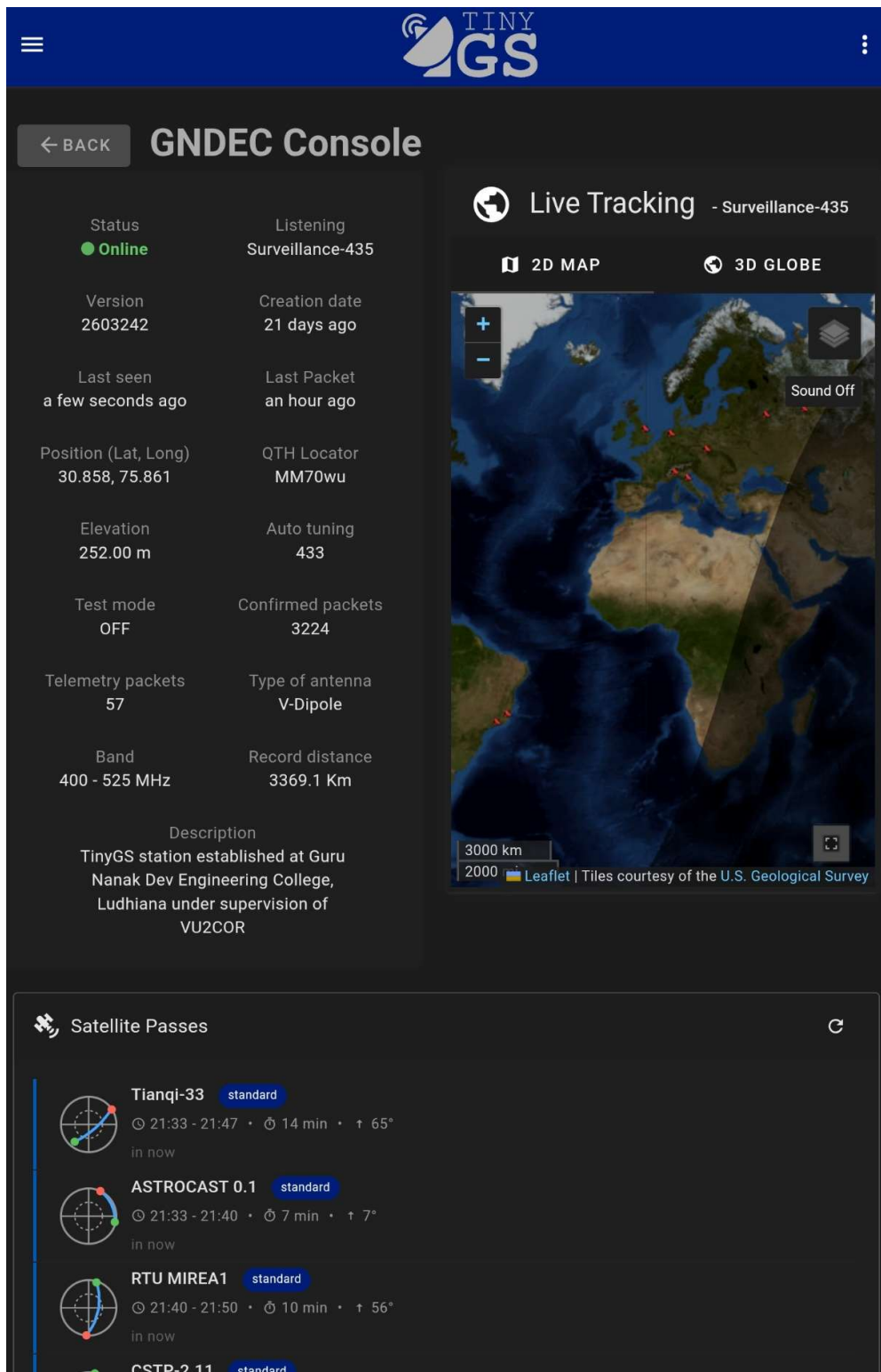


Fig.15 Software Dashboard



Fig.16 Software Dashboard

During verification, the system is tested under different signal conditions to ensure stability and accuracy. Factors such as signal strength, data consistency, and response time are observed carefully. Any issues such as signal loss, incorrect data decoding, or display errors are identified and resolved by debugging or improving antenna positioning and circuit connections. This process ensures that the final system operates reliably and provides accurate real-time satellite data monitoring.

CHAPTER 4

CONCLUSION AND FUTURE SCOPE

4.1 Conclusion

The LoRa-based Satellite Ground Station project has been successfully designed and developed using ESP32, LoRa module, OLED display, and supporting components. This project demonstrates the practical implementation of wireless communication and embedded systems for real-time satellite data reception. Through this system, LoRa signals transmitted by satellites are received, decoded, and displayed effectively.

This project not only works as a functional ground station but also enhances understanding of IoT communication, long-range wireless transmission, and real-time data processing. It is cost-effective, easy to implement, and useful for both learning and practical applications. Overall, the project meets its objectives and clearly shows how embedded systems can be used in satellite communication and tracking applications.

4.2 Future Scope

The project can be further improved by adding more advanced features in the future.

Some possible future scope areas are:

- Integration with cloud platforms (e.g., TinyGS) for global data sharing
- Use of high-gain antennas to improve signal range and reception quality
- Adding GPS module for precise ground station location tracking
- Implementation of automatic satellite tracking system
- Development of mobile or web-based interface for remote monitoring
- Use of advanced microcontrollers for handling complex data processing
- Integration with multiple satellite frequencies for broader coverage
- Data logging and storage for analysis and research purposes

In the future, the project can be expanded into a fully automated and intelligent satellite ground station system with enhanced accuracy, connectivity, and real-time monitoring capabilities.

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